

2016.0 RANGE ROVER (LG), 206-06

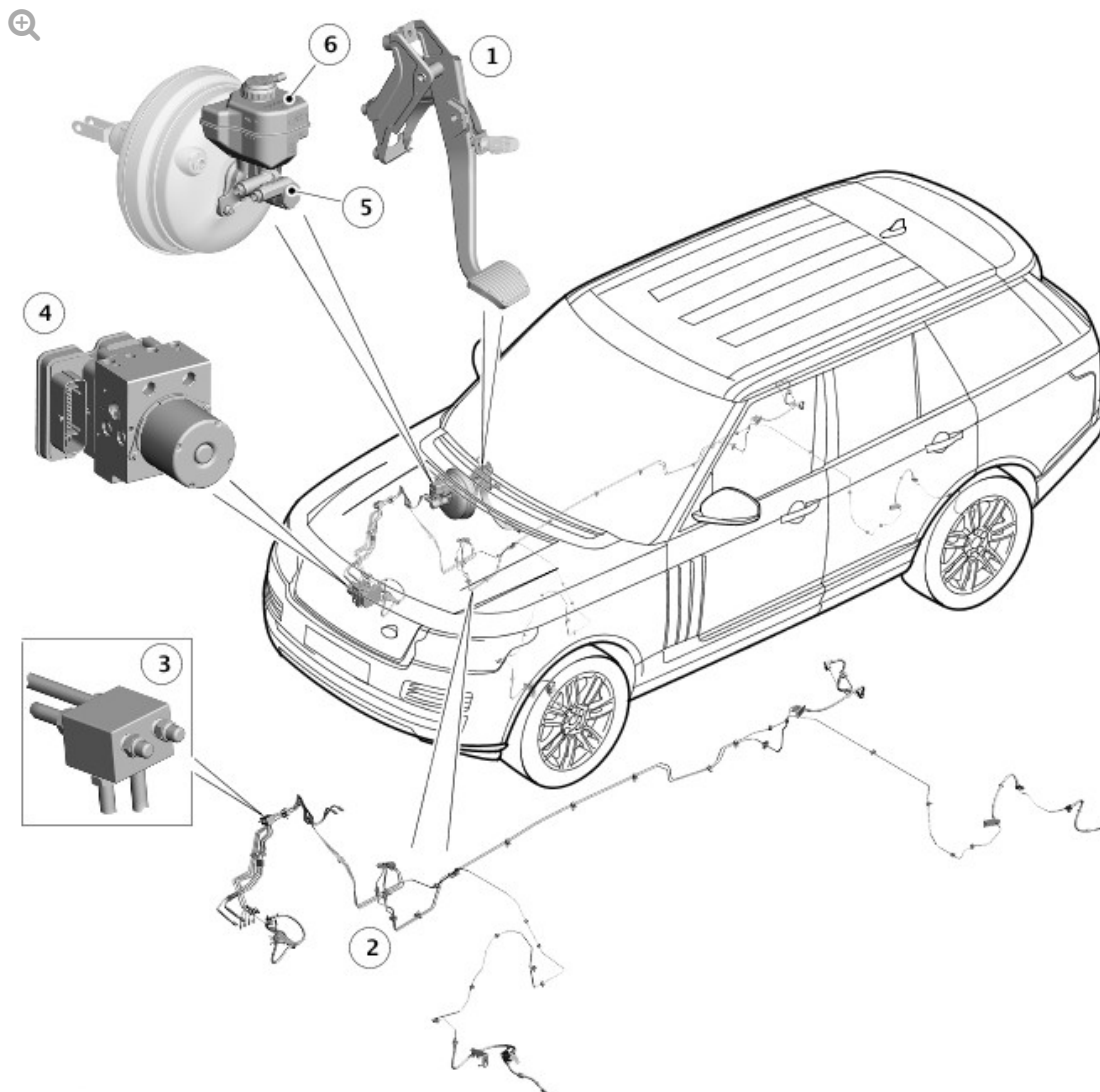
HYDRAULIC BRAKE ACTUATION

DESCRIPTION AND OPERATION

COMPONENT LOCATION

NOTE:

RHD (right-hand drive) installation shown, LHD (left-hand drive) installation similar.



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ITEM	DESCRIPTION
1	Brake pedal
2	Brake pipes and hoses
3	Bleed block
4	Anti-lock brake system control module
5	Brake master cylinder
6	Brake fluid reservoir

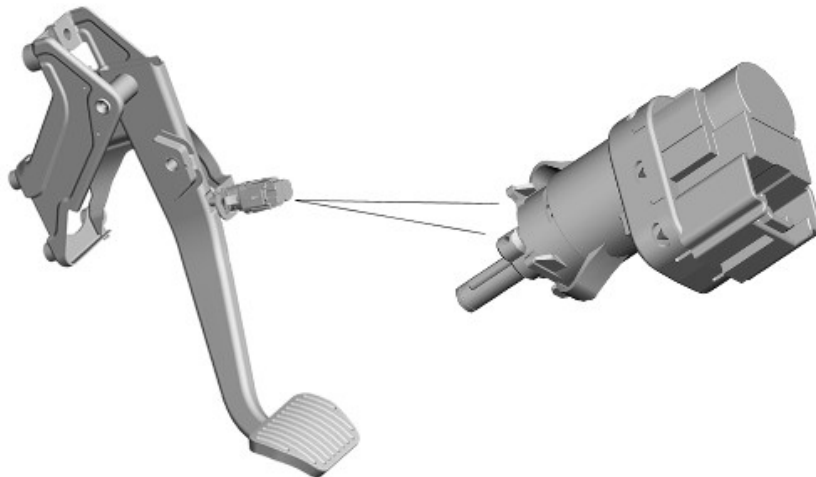
OVERVIEW

The hydraulic brake actuation system consists of the brake pedal, the brake

master cylinder and fluid reservoir, the HCU (hydraulic control unit) and the brake pipes and hoses. The brake pipes and hoses are arranged to provide a front/rear split hydraulic system.

DESCRIPTION

BRAKE PEDAL



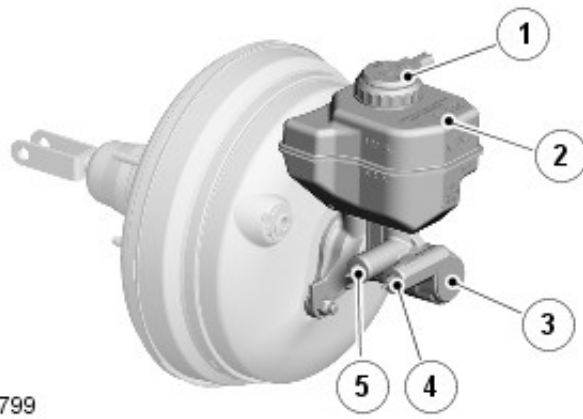
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The brake pedal is installed in a bracket attached to the rear side of the engine bulkhead. A clevis pin and clip connect the brake pedal to the push rod of the brake booster.

The brake pedal switch is installed in the bracket and operated by movement of the brake pedal.

For additional information, refer to: Braking Control System (206-11 Brake Controls, Description and Operation).

BRAKE MASTER CYLINDER AND FLUID RESERVOIR



E141799

ITEM	DESCRIPTION
1	Filler cap and level sensor
2	Brake fluid reservoir
3	Brake master cylinder
4	Secondary circuit connection
5	Primary circuit connection

The brake master cylinder and fluid reservoir are attached to the front of the brake booster, on the driver's side of the engine compartment. The brake master cylinder is a tandem design that supplies pressure to two independent (primary and secondary) hydraulic circuits for the front and rear brakes respectively. If a failure occurs in one hydraulic circuit, the remaining circuit will still operate effectively, although brake pedal travel and vehicle braking distances will increase.

The fluid reservoir is installed on top of the master cylinder and retains sufficient volume of hydraulic fluid to allow for normal system use and to compensate for the replenishment of the system as the brake pads wear.

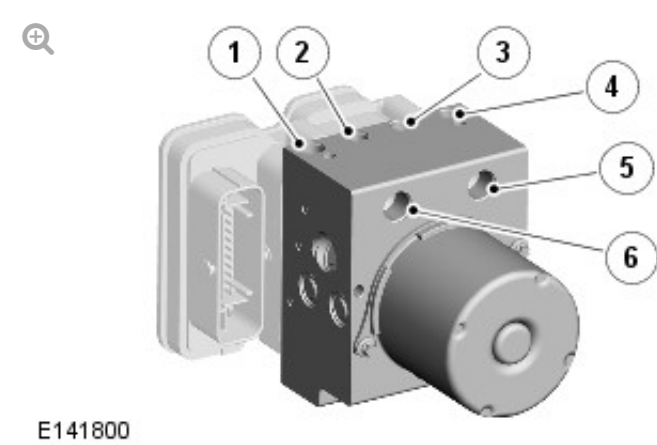
A brake fluid level sensor is incorporated into the cap of the fluid reservoir.

BRAKE PIPES AND HOSES

Preformed rigid brake pipes distribute pressure from the master cylinder to the four corners of the vehicle via the HCU. Braided steel hoses connect the

brake pipes to the front and rear calipers. For service use, a bleed block is incorporated into the two brake pipes between the master cylinder and the HCU.

HYDRAULIC CONTROL UNIT



ITEM	DESCRIPTION
1	Rear right brake connection
2	Rear left brake connection
3	Front right brake connection
4	Front left brake connection
5	Primary circuit connection
6	Secondary circuit connection

The HCU is an integral part of the ABS (anti-lock brake system) control module, which is installed in the front corner of the engine compartment, on the driver side. The ABS control module is attached to a bracket installed between the front bumper mounting bracket and the suspension housing.

The HCU is a four channel unit that modulates the supply of hydraulic pressure to the brakes under the control of the ABS control module. The primary and secondary outlet ports of the master cylinder are connected to primary and secondary inlet ports of the HCU. The primary circuit in the HCU supplies hydraulic pressure to two separate outlet ports for the front

brakes. The secondary circuit provides hydraulic pressure to two separate outlet ports for the rear brakes.

The HCU has three brake operating modes: Normal braking / EBD (electronic brake force distribution); ABS braking; Active braking. Details of these operating modes are given in the Brakes Control section of this Manual.

For additional information, refer to: Braking Control System (206-11 Brake Controls, Description and Operation).

OPERATION

BRAKE FLUID LEVEL SENSOR

If the hydraulic fluid in the reservoir falls to a predetermined low level, the switch contacts close and provide a ground signal to the CJB (central junction box). The CJB then sends a high speed CAN (controller area network) signal to the instrument cluster and the ABS control module. On receipt of the signal:

- The instrument cluster illuminates the red brake warning indicator, and the message center displays the critical warning indicator (red warning triangle) and the message Brake Fluid Low.
- The ABS control module disables DSC (dynamic stability control) functions and sends a high speed CAN signal to the instrument cluster to continuously illuminate the DSC warning indicator.